Simon Fleet

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Profile

Software engineer working primarily in Python and TypeScript, building applications end-to-end from data model to production. First Class Master's in Mathematics.

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL
Backend and Data: FastAPI, PostgreSQL, SQLite, Supabase, Drizzle, data modelling
Mobile and Web: React Native, Expo, Streamlit
Mathematics: Linear algebra, measure-theoretic probability, stochastic processes, optimisation, numerical methods
Tools: Git, GitHub, EAS, pytest

Software Engineering Experience and Projects

Order Book Market Microstructure Simulator *Independent Project | GitHub* *2026 - Present*

- Built a Python agent based market simulator around a central limit order book, modelling interactions between configurable trading agents and market making strategies.
- Implemented price time priority matching for market and limit orders, including partial fills, cancellations and execution across multiple price levels using FIFO queues.
- Separated order book execution, agent strategy and settlement logic, with cash and inventory reserved against open orders to prevent agents from overcommitting their assets.
- Wrote pytest tests for matching, cancellations, budgeting and settlement, including long-run checks that total cash and inventory remain conserved across multi-agent simulations.

Window Quotation and Ordering Platform *Cofounder & Engineer* *Feb 2026 - Jul 2026*

- Developed a B2B platform for the window and door supply chain, with a tablet surveying application and supplier web portal sharing a Supabase/PostgreSQL backend.
- Created a web portal allowing suppliers and manufacturers to configure prices, product options, and compliance rules.
- Designed a nine-stage TypeScript BOM engine that reconstructs nested divider and frame topology, deriving accurate cut lengths, glass sizes, and hardware quantities across mixed fixed and opening sections.
- Built the pricing pipeline linking BOM output to supplier price lists, applying multi-level mark-ups, discounts, and VAT, then atomically freezing quotations for penny-accurate, consistent offline sync.

Farrier Fleet *Founder & Sole Engineer* *Dec 2025 - Present*

- Shipped an iOS app end-to-end for farriers to manage clients, horses, appointments, and records.
- Used regularly by $5+$ farriers in North Carolina, replacing paper-based record keeping and saving around 30 minutes of administration per week.
- Built an offline-first relational data model across more than 14 core entities, supporting structured long-term horse and client records.
- Managed the production release, including RevenueCat subscriptions, EAS builds, and the App Store review process.

Education and Research

University of Bristol *Mathematics MSci, First Class Honours* *2020 - 2024*

- Awarded the Howell Peregrine Project Prize for the best project in the cohort; achieved 91% in the research project and an 80% final-year module average.
- Authored *Short Sums of the Liouville Function over Function Fields*, proving square-root cancellation for the function-field analogue of the Liouville function.

Additional Experience

ML Data Analyst, DataAnnotation *July 2024 - September 2025*

- Designed and managed data-generation projects, producing thousands of high-quality human annotations to improve LLM performance on reasoning and software-engineering tasks.
- Analysed and cleaned generated data to remove ambiguity, improve consistency, and strengthen dataset quality.

Climbing Coach *September 2023 - March 2024*

- Coached young athletes preparing for national competitions, with one athlete selected for the England team.

Mathematics Tutor *2017 - 2023*

- Provided one-to-one GCSE and A-Level tutoring, helping students improve their confidence, mathematical reasoning, and examination performance.